A CASE-STUDY BASED REPORT

ON

CARBON EMISSION PREDICTION USING GRADIENT BOOSTING AND LINEAR REGRESSION

submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

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by

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Abstract

Machine Learning is an important branch of Artificial Intelligence that enables computers to learn patterns from data and make predictions without being explicitly programmed for every situation. This project focuses on developing a machine learning system that can analyze real-world data, identify meaningful patterns, and classify new observations into appropriate categories. The study includes dataset collection, data preprocessing, exploratory data analysis, algorithm selection, model training, and performance evaluation. Exploratory Data Analysis is used to understand the structure of the dataset, identify missing values, examine feature distributions, and discover relationships between variables. Suitable classification algorithms are considered based on the characteristics of the dataset and the requirements of the problem. The project also emphasizes explainability, which helps users understand the factors influencing a model's predictions. Model performance can be evaluated using metrics such as accuracy, precision, recall, and F1-score. The overall objective is to demonstrate how machine learning techniques can support data-driven decision-making while recognizing the limitations of automated predictions and the importance of responsible data usage.

1. Introduction

1.1 Background

Machine Learning is a technology that allows computer systems to learn from historical data and use the knowledge gained to make predictions or decisions. Traditional programming requires developers to define explicit rules, whereas machine learning algorithms identify patterns and relationships from examples.

Machine learning is widely applied in healthcare, agriculture, education, finance, cybersecurity, and business analytics. For example, classification models can categorize customer feedback, detect suspicious transactions, identify plant diseases, or predict whether a particular condition is present based on collected data.

However, a model's prediction is useful only when it is sufficiently reliable and its limitations are understood. A model may produce incorrect results if the dataset contains errors, insufficient examples, biased observations, or features that do not represent real-world conditions. Therefore, a complete machine learning project must include data analysis, proper model evaluation, and an understanding of how predictions are generated.

This project explores a systematic machine learning workflow that combines data preparation, exploratory analysis, classification, and explainability to develop a better understanding of predictive modelling.

1.2 Motivation for the Project

The increasing availability of digital data has created opportunities to solve practical problems using machine learning. However, raw data alone cannot provide meaningful conclusions unless it is properly analyzed and processed.

The motivation for this project is to understand how a machine learning model can transform raw data into useful predictions. Another important motivation is to understand why a model produces a particular output rather than treating its prediction as automatically correct.

The project provides practical experience in handling datasets, identifying important features, selecting suitable algorithms, training models, and evaluating their performance. It also highlights the importance of transparency, fairness, and responsible use when applying machine learning to real-world situations.

Through this work, the study aims to connect theoretical concepts learned in machine learning with their practical implementation in a programming environment.

1.3 Overview of the Work Carried Out

The project follows a sequence of stages to develop and evaluate a machine learning classification system.

First, a suitable dataset is identified and examined to understand its structure, attributes, and target variable. Next, data preprocessing is carried out to handle missing values, duplicate records, inconsistent data, and unsuitable feature formats where necessary.

Exploratory Data Analysis is then performed using descriptive statistics and visualizations to understand feature distributions and relationships. Based on the dataset and problem requirements, an appropriate classification algorithm is selected.

The dataset is divided into training and testing subsets to assess how well the model performs on previously unseen observations. The model is trained using the training data and evaluated using suitable performance metrics. Finally, the factors influencing the model's predictions are examined using appropriate interpretation techniques, where supported by the implementation.

The complete workflow helps demonstrate the relationship between data quality, algorithm selection, model performance, and the interpretability of machine learning predictions.

1.4 Scope and Responsible-Use Statement

The scope of this project is limited to studying a machine learning classification problem using a selected dataset. It includes data understanding, preprocessing, exploratory analysis, algorithm implementation, performance evaluation, and interpretation of model outputs.

The project is intended for academic learning and experimental analysis. Its results depend on the quality, size, and representativeness of the selected dataset. Performance observed during testing does not guarantee identical performance when the model is used with new data in a different environment.

Responsible use requires protecting sensitive information, avoiding unnecessary collection of personal data, identifying possible dataset biases, and communicating uncertainty clearly. Model predictions should not be treated as unquestionable facts or as substitutes for expert judgement in high-impact situations.

Any explanation generated by an interpretation technique should also be understood within its methodological limitations. The project therefore treats machine learning as a decision-support approach rather than a system that guarantees correct outcomes.

2. Problem Statement

2.1 Case-Study Scenario

In many real-world applications, large amounts of data are collected through digital systems, surveys, experiments, sensors, and other sources. Although this data contains useful information, manually examining every record can be time-consuming and may make it difficult to identify complex patterns.

Consider a case study in which a dataset contains several input features and a target category that must be predicted. The objective is to use historical examples to train a machine learning model that can classify new observations based on the relationships learned from the data.

The challenge is not limited to producing a prediction. The system should also be evaluated to determine whether its predictions are sufficiently reliable and, where possible, which features contribute to its decisions.

This scenario provides a foundation for studying how data analysis and machine learning can be combined to solve a classification problem in a structured manner.

2.2 Central Case-Study Question

The central question guiding this project is:

How can an appropriate machine learning classification algorithm be developed and evaluated to produce reliable predictions from real-world data while providing understandable explanations of its outputs?

To address this question, the study examines the following supporting questions:

1. **What patterns and relationships can be identified in the selected dataset?**

2. **What preprocessing steps are required before training the model?**

3. **Which classification algorithm is suitable for the dataset and its target variable?**

4. **How accurately can the model classify previously unseen observations?**

5. **How can the factors influencing the model's predictions be interpreted?**

6. **What limitations must be considered before applying the model in practice?**

2.3 Formal Problem Definition

Let the dataset be represented as:

\[D=\{(x_i,y_i)\}_{i=1}^{N} \]
where:

- **\(N\) represents the total number of observations.=**
- **\(x_i\) represents the input feature vector for observation \(i\).**
- **\(y_i\) represents the actual class label associated with that observation.**

The input vector may contain numerical or categorical attributes, depending on the selected dataset. The target variable represents the category that the model is expected to predict.

The objective is to learn a classification function:

\[f:X\rightarrow Y \]
where \(X\) is the input feature space and \(Y\) is the set of possible target classes.

The trained function \(f\) receives an input observation and produces a predicted class label. Its performance is assessed by comparing predicted labels with actual labels from a separate test dataset.

In addition to predictive performance, the project considers whether the model's outputs can be interpreted meaningfully. The final model should be evaluated using suitable metrics and should acknowledge uncertainty, data limitations, and possible sources of error.

2.4 Objectives that Guide the Study

The main objectives of the project are:

To understand the structure, attributes, and target variable of the selected dataset.

To identify missing values, duplicate records, unusual observations, and inconsistencies.

To perform Exploratory Data Analysis to discover patterns and relationships among features.

To prepare the dataset using suitable preprocessing and feature-transformation techniques.

To select and implement an appropriate machine learning classification algorithm.

To train the model using a training dataset and evaluate it using unseen test observations.

To measure performance using accuracy, precision, recall, F1-score, and a confusion matrix, where appropriate.

To examine the contribution of relevant features to the model's predictions using suitable interpretation methods.

To identify the limitations of the model and discuss responsible approaches to its potential use.

2.5 Assumptions and Limitations

The project is based on the assumption that the selected dataset contains sufficient information to investigate the chosen classification problem. It is also assumed that the target labels are meaningful and that the input features are relevant to the prediction task.

Where the dataset contains labelled examples, supervised learning can be used to learn the relationship between the input features and the target variable. The training and testing procedure is assumed to provide a reasonable evaluation when the data split is appropriate and no information from the test set leaks into model training.

Several limitations may affect the results. The dataset may be small, imbalanced, incomplete, outdated, or insufficiently representative of the population or conditions in which the model might eventually be used. The model may also perform differently on data collected from another source.

Furthermore, high accuracy alone does not establish that a model is fair, well-calibrated, or suitable for every application. Explainability techniques may provide useful insights, but their explanations are not necessarily complete descriptions of the model's internal behaviour.

The conclusions of the project must therefore remain within the limits of the available data and the experiments actually performed.

3. Dataset and Exploratory Data Analysis

3.1 About the Dataset

A dataset is a structured collection of observations used to identify patterns and train machine learning models. In a classification project, each observation contains a set of input features and, in most supervised learning tasks, a target label representing the category to be predicted.

The dataset selected for this project should be relevant to the chosen case study and should contain sufficient observations to support meaningful analysis. It may be obtained from a publicly available repository, an open research dataset, or a permitted data collection process.

The dataset is examined to understand its size, number of features, data types, target categories, and overall quality. Numerical features may represent measurable quantities, while categorical features represent groups or labels. The target variable determines the classification task the model must perform.

Before using the dataset for training, its suitability must be checked. This includes examining missing values, duplicate records, inconsistent entries, class imbalance, and the relevance of the available features.

The dataset's source, number of rows and columns, feature descriptions, and target variable should be documented to make the experiment easier to understand and reproduce.

Details to include after selecting your dataset:

7. **Dataset name and source.**

8. **Total number of records and features.**

9. **Input features and their meanings.**

10. **Target variable and its possible classes.**

11. **Data types and dataset format.**

12. **Any known limitations of the collected data.**

3.2 Purpose of Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) is the process of examining and summarizing a dataset before developing a machine learning model. Its main purpose is to understand the characteristics of the data and identify issues that could affect the quality of the predictions.

EDA helps reveal the distribution of individual features, relationships between variables, differences between target classes, and possible unusual observations. It also helps determine which preprocessing operations may be required before training.

Descriptive statistics such as mean, median, standard deviation, minimum, and maximum values provide an overview of numerical attributes. Frequency counts help explain the distribution of categorical variables. Visualizations such as histograms, bar charts, box plots, and correlation heatmaps make patterns easier to recognize.

EDA also helps identify whether the target classes are balanced. If one class contains substantially more observations than another, a model may perform well on the majority class while failing to identify minority-class examples.

The findings from EDA guide decisions about preprocessing, feature selection, algorithm choice, and model evaluation. Therefore, EDA is an essential stage in developing a reliable machine learning workflow.

3.3 Preparation Steps Actually Performed

Data preparation converts the original dataset into a suitable format for machine learning. The exact operations depend on the dataset's structure and the requirements of the selected algorithm.

The preparation process may include the following steps:

1. Data loading: The dataset is imported into the programming environment using an appropriate library, such as Pandas in Python.

2. Data inspection: The number of records, column names, data types, and sample observations are examined to understand the structure of the dataset.

3. Missing-value analysis: Columns containing missing values are identified. Depending on the context, missing entries may be removed or filled using appropriate methods such as median, mean, or mode imputation.

4. Duplicate checking: Repeated records are identified and handled when they represent unintended duplication.

5. Data-type correction: Features are converted into suitable numerical, categorical, or date formats where necessary.

6. Categorical encoding: Categorical input features are transformed into numerical representations if required by the selected algorithm.

7. Feature scaling: Numerical features are standardized or normalized when the chosen algorithm benefits from comparable feature scales.

8. Feature and target separation: The input variables are separated from the target variable to define the prediction task.

9. Training and testing split: The prepared dataset is divided into training and testing subsets. The training data is used to fit the model, while the test data is reserved for evaluating its performance on unseen observations.

10. Data leakage prevention: Preprocessing parameters are learned from the training data and then applied to the test data. This prevents information from the test set from improperly influencing model training.

Only the steps actually performed in the implementation should be reported as completed. If a dataset does not contain missing values, for example, missing-value imputation should not be claimed as an operation that was carried out.

3.4 EDA Exploration – Implementation

EDA can be implemented using Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn. These libraries support data inspection, statistical calculations, and visualization.

The initial stage involves loading the dataset and displaying a sample of its observations. The dimensions of the dataset are then examined to determine the number of rows and columns. Data types and missing-value counts are checked to identify potential preprocessing requirements.

Next, descriptive statistics are calculated to understand the distribution of numerical features. Frequency distributions are used to inspect categorical features and determine the number of observations in each target class.

Histograms can be used to examine the distribution of numerical attributes, while bar charts can compare the frequency of categorical values. Box plots can help identify unusual observations that may require investigation. A correlation heatmap can summarize linear relationships between numerical features, although correlation alone does not establish causation.

The target variable is also examined to determine whether the classification problem has balanced or imbalanced classes. Where relevant, relationships between input features and the target are investigated to identify potentially informative patterns.

The EDA findings are documented through tables, plots, and brief explanations. These outputs provide a foundation for selecting suitable preprocessing methods and machine learning algorithms.

3.5 Summary of EDA Findings

The results of Exploratory Data Analysis are summarized to explain the main characteristics of the dataset and the decisions made during data preparation.

The summary should describe the dataset's size, important features, target-class distribution, missing-value patterns, duplicate records, and notable relationships between variables. It should also explain whether the data required cleaning, encoding, scaling, or other transformations.

For example, if EDA reveals that a numerical feature contains a highly skewed distribution, the report may discuss whether that distribution affects the selected algorithm. If the target classes are imbalanced, the report may explain why precision, recall, and F1-score are important in addition to accuracy.

The summary should distinguish between observations supported by the analysis and assumptions that have not been verified. Graphs and numerical statistics should be included wherever possible to support the conclusions.

The findings from this stage are used to guide algorithm selection and establish an understanding of the dataset before model training. Actual observations and numerical results should be inserted after running the EDA implementation.

4. Algorithm Selection and Implementation

Algorithm selection is an important stage in a machine learning project because different algorithms learn patterns in different ways. The most suitable algorithm depends on the dataset size, feature types, relationships between variables, number of target classes, computational requirements, and need for interpretability.

For a classification problem, possible algorithms include Logistic Regression, Decision Tree, Random Forest, Support Vector Machine, and K-Nearest Neighbors. These algorithms differ in complexity, predictive behaviour, and interpretability.

A Decision Tree is relatively easy to understand because it makes predictions through a sequence of feature-based rules. Random Forest combines predictions from multiple decision trees and can model complex relationships. Logistic Regression provides a useful baseline for classification problems and can be comparatively easy to interpret. Support Vector Machine may be effective for datasets with complex class boundaries, while K-Nearest Neighbors classifies observations based on nearby examples.

The selected algorithm should be justified according to the actual dataset and project requirements rather than chosen solely because it is popular.

4.1 Model Implementation Process

The implementation begins by defining the input features and target variable. The dataset is then divided into training and testing subsets using an appropriate split, such as 80% for training and 20% for testing, if that split suits the dataset.

The selected algorithm is initialized using a suitable programming library. In Python, Scikit-learn provides implementations of many common classification algorithms and evaluation metrics.

The model is trained using the training subset. During this process, the algorithm learns patterns that connect the input features with the target classes. After training, the model generates predictions for the test subset, which contains observations that were not used to fit the model.

The predicted labels are compared with the actual labels to evaluate performance. If model parameters are tuned, that process should use training data and an appropriate validation strategy rather than repeatedly optimizing against the test set.

4.2 Model Evaluation Metrics

The following metrics can be used to assess the classification model.

Accuracy: The proportion of all predictions that are correct.

$$\text{Accuracy}=\frac{TP+TN}{TP+TN+FP+FN}$$

Precision: The proportion of positive predictions that are correct.

$$\text{Precision}=\frac{TP}{TP+FP}$$

Recall: The proportion of actual positive cases correctly identified.

$$\text{Recall}=\frac{TP}{TP+FN}$$

F1-score: The harmonic mean of precision and recall.

$$F1=2\times\frac{\text{Precision}\times\text{Recall}}{\text{Precision}+\text{Recall}}$$

Here, TP represents true positives, TN true negatives, FP false positives, and FN false negatives for a binary classification task.

A confusion matrix provides a detailed comparison of actual and predicted classes. For multiclass classification, the metrics can be calculated separately for each class and summarized using macro or weighted averages.

Accuracy may be sufficient for some balanced datasets, but it can be misleading when the classes are imbalanced. Therefore, the choice of evaluation metrics should reflect the purpose and risks of the classification task.

4.3 Explainability of the Model

Explainability refers to the ability to understand the factors that influence a model's predictions. It is particularly useful when users need to investigate why the model assigned a particular class to an observation.

The explanation method depends on the selected algorithm. For a Decision Tree, the prediction path can be examined to identify the rules used to classify an observation. For a Random Forest, feature-importance measures can help identify features that contribute to predictive performance, although importance values do not necessarily explain individual predictions or establish causation.

Model-agnostic methods such as SHAP or LIME may be used to investigate individual predictions when appropriate. These methods estimate how features contribute to a model's output, but their explanations have assumptions and limitations that should be acknowledged.

Explainability is therefore used as a complementary tool alongside quantitative performance evaluation. A model should not be considered reliable merely because its predictions can be explained.

4.4 Results and Discussion

After model training and testing, the evaluation results should be presented in a table.

Evaluation metric	Obtained result
Accuracy	[Enter measured value]
Precision	[Enter measured value]
Recall	[Enter measured value]
F1-score	[Enter measured value]

The results should be interpreted in relation to the dataset and the original problem statement. For example, a high accuracy value indicates that many test observations were classified correctly, but it does not guarantee that every target class was identified equally well.

Precision and recall should be examined to understand the types of classification errors made by the model. The confusion matrix can help identify which classes are frequently confused.

If multiple algorithms are implemented, their results can be compared using the same data split and evaluation procedure. This helps determine which model performs better under the chosen experimental conditions.

The discussion must report the actual measured results rather than assumed or fabricated values. If explainability methods are used, their findings should also be discussed alongside the model's limitations.

5. Conclusion

This project presents a structured approach to developing and evaluating an explainable machine learning classification system. It covers the major stages of a machine learning workflow, including dataset understanding, data preprocessing, Exploratory Data Analysis, algorithm selection, model training, performance evaluation, and interpretation of predictions.

EDA helps identify important characteristics of the dataset and supports decisions about data preparation. Algorithm selection determines how the model learns relationships between input features and target classes. Evaluation metrics provide a quantitative assessment of predictive performance, while explainability techniques help investigate the factors influencing model outputs.

The project also highlights the importance of data quality, appropriate evaluation procedures, and responsible use. A machine learning model may perform well on a particular test dataset but still encounter difficulties when applied to different data or real-world conditions. Therefore, the model's results must be interpreted within the limits of the available evidence.

Overall, this study provides practical insight into how machine learning can be used to analyze data and generate predictions in a systematic manner. It also demonstrates that successful machine learning development involves more than training an algorithm: it requires careful data analysis, meaningful evaluation, transparent reporting, and awareness of limitations.

Future improvements may include evaluating additional algorithms, expanding the dataset, improving feature selection, tuning model parameters, testing the model on independent datasets, and applying suitable explainability techniques. These improvements can help determine whether the system generalizes effectively and whether its predictions are useful for the intended application.